Investigation: RECRUSUL S.A. (CVM code 12572), fiscal year 2015, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 1, filed on 2023-05-08): RCSL3 (Ações Ordinárias, Bolsa, since 1985-12-20); RCSL4 (Ações Preferenciais, Bolsa, since 1985-12-20).

Document used

2015 DFP, version 1, filed on 2016-08-15; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2016-08-15).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=59130

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −6.164 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 8 (n = 18)
- Percentile in group and year: 3 (n = 376)
- The company's own history: median −2.749; Tukey range (1.5 × IQR): from −10.148 to 2.734; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.000 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 21)
- Percentile in group and year: 1 (n = 460)
- The company's own history: median 1.019; Tukey range (1.5 × IQR): from 0.016 to 2.545; value below the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.591 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 12 (n = 21)
- Percentile in group and year: 10 (n = 481)
- The company's own history: median 1.024; Tukey range (1.5 × IQR): from 0.670 to 1.547; value below the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.055 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 55 (n = 21)
- Percentile in group and year: 67 (n = 464)
- The company's own history: median 0.959; Tukey range (1.5 × IQR): from 0.364 to 1.661; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 0.288 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 7 (n = 21)
Percentile in group and year: 2 (n = 483)
The company's own history: median 1.277; Tukey range (1.5 × IQR): from −1.561 to 3.290; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 1.327 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 92 (n = 18)
Percentile in group and year: 94 (n = 393)
The company's own history: median 0.971; Tukey range (1.5 × IQR): from 0.403 to 1.373; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.726 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 12 (n = 21)
Percentile in group and year: 9 (n = 470)
The company's own history: median 0.904; Tukey range (1.5 × IQR): from −2.288 to 6.120; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.417 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 12 (n = 21)
Percentile in group and year: 4 (n = 489)
The company's own history: median −0.044; Tukey range (1.5 × IQR): from −0.930 to 0.525; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.203 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 82 (n = 484)
The company's own history: median 1.160; Tukey range (1.5 × IQR): from −0.527 to 3.223; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: 4,041.1 days
Alert by the study 1 rule: yes
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 42.4 days)
Percentile in sector and year: 98 (n = 21)
Percentile in group and year: 99 (n = 449)
The company's own history: median −31.7 days; Tukey range (1.5 × IQR): from −489.6 days to 693.8 days; value above the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix

(decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 64.01 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 92 (n = 438)
The company's own history: median 125.18 p.p.; Tukey range (1.5 × IQR): from 90.36 p.p. to 147.00 p.p.; value below the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.406 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 12 (n = 21)
Percentile in group and year: 4 (n = 488)
The company's own history: median −0.045; Tukey range (1.5 × IQR): from −0.848 to 0.475; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −32.948 (index)
Components: X1 = −2.169; X2 = −5.191; X3 = −0.151; X4 = −0.747 (indices)
Percentile in sector and year: 7 (n = 22)
The company's own history: median −7.471; Tukey range (1.5 × IQR): from −14.109 to −2.420; value below the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 1
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-29
- DFP 2011, v1: 2012-03-30
- DFP 2012, v1: 2013-03-27
- DFP 2012, v2: 2013-04-04
- DFP 2013, v1: 2014-03-31
- FCA 2015, v1: 2015-03-03
- FCA 2015, v2: 2015-05-15
- DFP 2014, v1: 2015-06-30
- FCA 2015, v3: 2015-09-03
- DFP 2015, v1: 2016-08-15
- DFP 2016, v1: 2017-03-31

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: Taticca Auditores Independentes SS.
Auditor in the 2022 FCA: Taticca Auditores Independentes SS.
Auditor in the 2023 FCA: Taticca Auditores Independentes SS.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: DRS Auditores -> Michelon & Cia Auditores e Consultores SS. The outgoing firm audited 5 consecutive fiscal years (2007 to 2011), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2015: Michelon & Cia Auditores e Consultores SS -> Baker Tilly Brasil RS Auditores Independentes Sociedade Simples. The outgoing firm audited 3 consecutive fiscal years (2012 to 2014), by the start and end dates in the FCA.
- 2020: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples -> Taticca Auditores Independentes SS. The outgoing firm audited 5 consecutive fiscal years (2015 to 2019), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful

accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:40:45+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.